

第 11 题
2021010697



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$$w_1 \|x - \bar{u}_1\|^2 = w_2 \|x - \bar{u}_2\|^2$$

$$\frac{1}{1+b} = \frac{1}{b+1}$$

1.1

$$\int_{-\infty}^{+\infty} p(x|w_1) dx = \frac{1}{\pi b} \int_{-\infty}^{+\infty} \frac{1}{1 + (\frac{x-a_1}{b})^2} dx$$

$$= \frac{1}{\pi} \int_{-\infty}^{+\infty} \frac{1}{1 + \theta^2} d\theta$$

$$= \frac{1}{\pi} \arctan \theta \Big|_{-\infty}^{+\infty}$$

$$= 1$$

1.2

$$p(w_1|x) = \frac{p(x|w_1)p(w_1)}{p(x|w_1)p(w_1) + p(x|w_2)p(w_2)}$$

$$= \frac{p(x|w_1)}{p(x|w_1) + p(x|w_2)}$$

∴ 当 $x = \frac{a_1+a_2}{2}$ 时 $p(x|w_1) = p(x|w_2) = \frac{1}{\pi b} \cdot \frac{1}{1 + (\frac{0}{2b})^2}$

有 $p(w_1|x) = p(w_2|x) = 0.5$

1.3

见图附件。

$$p(x|w_1) = p(x|w_2)$$

$$p(w_1|x) = \frac{p(x|w_1)}{p(x|w_1) + p(x|w_2)}$$

$$= \frac{1 + (x-5)^2}{1 + (x-3)^2 + 1 + (x-5)^2} I_x[-\infty, +\infty]$$

$$= \frac{x^2 - 10x + 26}{2x^2 - 16x + 36}$$

1.4

$$\lim_{x \rightarrow +\infty} p(w_1|x) = \lim_{x \rightarrow +\infty} \frac{\frac{1}{1 + (\frac{x-a_1}{b})^2}}{\frac{1}{1 + (\frac{x-a_1}{b})^2} + \frac{1}{1 + (\frac{x-a_2}{b})^2}}$$

$$= \lim_{x \rightarrow +\infty} \frac{1 + \frac{1}{b^2} x^2 - 2\frac{a_1}{b} x + \frac{a_1^2}{b^2}}{1 + \frac{1}{b^2} x^2 - 2\frac{a_1}{b} x + \frac{a_1^2}{b^2} + 1 + \frac{1}{b^2} x^2 - 2\frac{a_2}{b} x + \frac{a_2^2}{b^2}}$$

$$= 0.5$$

同理 $\lim_{x \rightarrow +\infty} p(w_2|x) = 0.5$ ∴ $\lim_{x \rightarrow +\infty} p(w_1|x) = \lim_{x \rightarrow +\infty} p(w_2|x) = 0.5$

2.1

$$p(x|w_1) = \frac{1}{2\pi} \exp(-\frac{1}{2} \vec{x}^T \vec{x})$$

$$p(x|w_2) = \frac{1}{2\pi} \exp(-\frac{1}{2} (\vec{x} - (1,1))^T (\vec{x} - (1,1)))$$

$$p(x|w_1) = p(x|w_2) \Leftrightarrow (\vec{x} - \begin{pmatrix} 1 \\ 1 \end{pmatrix})^T (\vec{x} - \begin{pmatrix} 1 \\ 1 \end{pmatrix}) = 0$$

$x_1 + x_2 = 1$ 均值点 $(0,0)$ 和 $(1,1)$

2.2

$$0.2 p(x|w_1) = 0.8 p(x|w_2)$$

$$\Leftrightarrow \vec{x}_0 = \frac{1}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \frac{1}{2} \ln \frac{1}{4} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$= \frac{1}{2} \begin{pmatrix} 1 - \ln 4 \\ 1 - \ln 4 \end{pmatrix}$$

决策面方程为 $(1,1)^T (\vec{x} - \frac{1}{2} \begin{pmatrix} 1 - \ln 4 \\ 1 - \ln 4 \end{pmatrix}) = 0$

2.3

2.1

$$g_1(x) = \vec{z}^T \vec{u}_1^T \vec{x} - \frac{1}{2} \vec{u}_1^T \vec{z}^T \vec{u}_1 + \ln p(w_1)$$

$$= \ln 0.5$$

$$g_2(x) = \vec{z}^T \vec{u}_2^T \vec{x} - \frac{1}{2} \vec{u}_2^T \vec{z}^T \vec{u}_2 + \ln p(w_2)$$

$$= \begin{pmatrix} 2 \\ 1 \end{pmatrix}^T \begin{pmatrix} 1 \\ 1 \end{pmatrix} \vec{x} - \frac{1}{2} (1,1) \begin{bmatrix} 2 \\ 1 \end{bmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \ln 0.5$$

$$= \begin{pmatrix} 1 \\ 0 \end{pmatrix}^T \vec{x} - \frac{1}{2} + \ln 0.5$$

$$g_1(x) = g_2(x) \Leftrightarrow x_1 - \frac{1}{2} = 0$$

2.4

$$g_1(x) = -x^T \frac{1}{2} \begin{pmatrix} 2 \\ 0 \end{pmatrix} \vec{x} + \begin{pmatrix} 2 \\ 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \end{pmatrix} - \frac{1}{2} \ln 2 + \ln 0.5$$

$$g_2(x) = -x^T \frac{1}{2} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \vec{x} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} - \frac{1}{2} (1,1) \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \ln 0.5$$

决策面为 $-x^T \frac{1}{2} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \vec{x} + \begin{pmatrix} -1 \\ -2 \end{pmatrix} \vec{x} + \frac{1}{2} 3 = 0$

$$0 = -x_1^2 + 0x_2^2 - 2x_1 - 4x_2 + 3 \Leftrightarrow (x_1+1)^2 + (x_2-2)^2 = 0$$

3.1

$$R_{\text{rand}} = E_x R(\alpha(x)|x)$$

$$= E_x (E_{w|x} (1 - P(w|x)))$$

$$= E_x (1 - \sum_{i=1}^n P(w_i|x))$$

3.2 $R_{\text{Bayes}} = E_x R(\alpha(x)|x)$

$$= E_x (1 - \max_{w_i} P(w_i|x))$$

对 $\forall x$, 有 $1 - \sum_{i=1}^n P(w_i|x) \geq 1 - \max_{w_i} P(w_i|x)$ ①

$\therefore R_{\text{Bayes}} \leq R_{\text{rand}}$

1.3 对式①, 当 $P(w_i|x) \equiv \frac{1}{n} \forall x$ 时, 不等式取等
即当后验分布为均匀分布时, 有 $R_{\text{Bayes}} = R_{\text{rand}}$.